

Family-Wide Collagen Variant Profiling Reveals Glycine Hotspots and Structurally Vulnerable ECM Genes Relevant to hEDS

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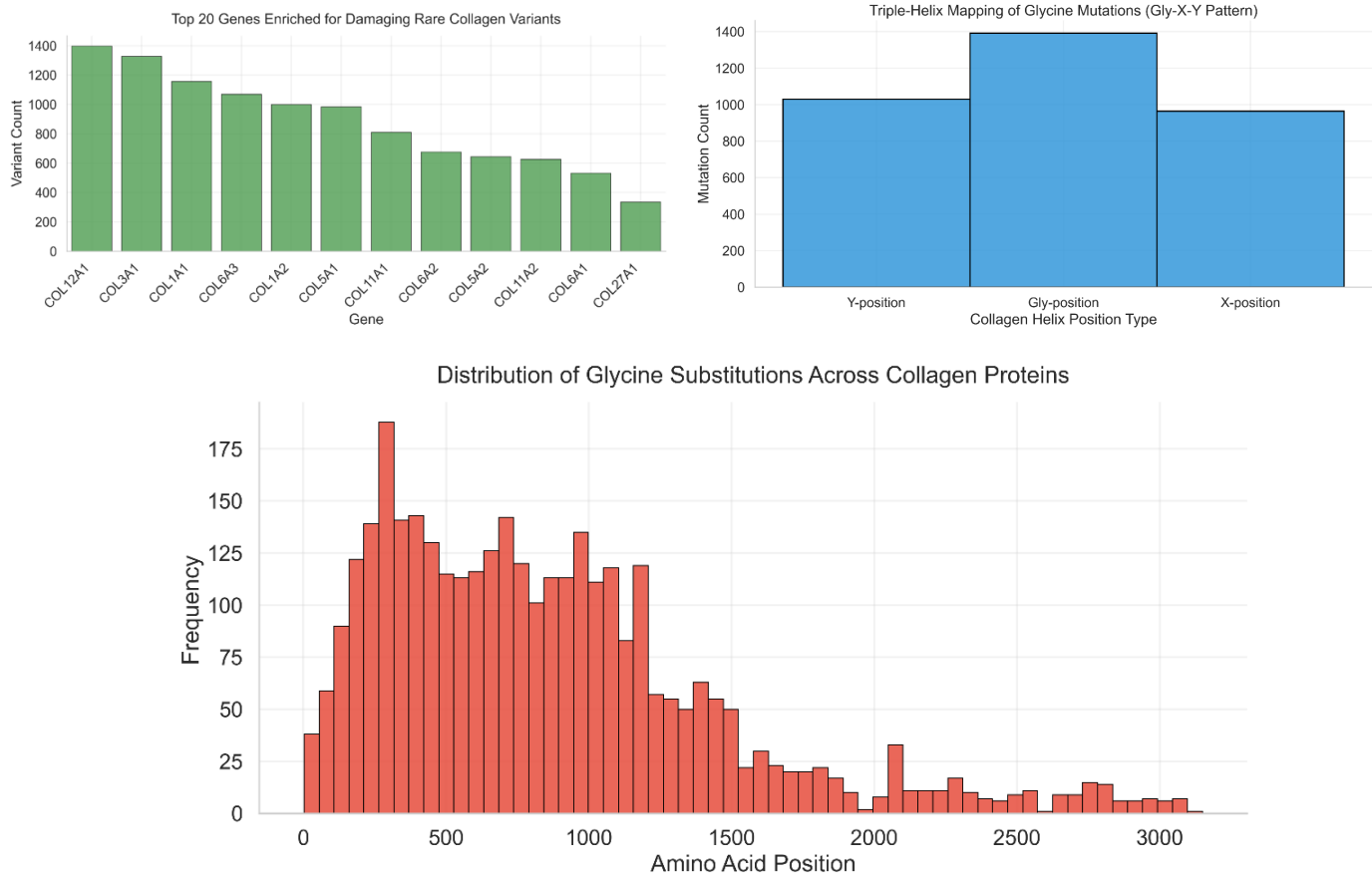
Abstract

Developed an end-to-end genomic analysis pipeline integrating ClinVar, Ensembl VEP, gnomAD, dbNSFP, and Mutalyzer to annotate over 35,000 variants across all collagen genes and process **1 TB of biological annotation data**. The pipeline performs automated HGVS normalization, large-scale functional annotation, and pathogenicity scoring (REVEL, CADD, PolyPhen, SIFT). Key outputs include **collagen-family glycine hotspot mapping, variant-burden profiling** across collagen types, and prioritization of structurally vulnerable ECM candidate genes such as **COL12A1** and **COL6A3**. Findings support a polygenic connective-tissue model relevant to the biology of hypermobile Ehlers-Danlos syndrome (hEDS).

Visualizations

I analyzed the master dataset generated by the pipeline, containing **35,788 collagen-gene variants**.

- Figure 1 (left): Top collagen genes enriched for damaging rare variants.
- Figure 2 (right): Triple-helix mapping of glycine substitutions (Gly-X-Y pattern), showing strong enrichment at the Gly position.
- Figure 3 (bottom): Distribution of glycine substitutions across collagen proteins shows non-random peaks.



Key Insights

- **COL12A1** and **COL6A3** emerge as leading ECM candidates supporting a **polygenic model** for hEDS.
- High-burden genes show **domain-level structural vulnerability**, with family-wide patterns consistent with **triple-helix constraints**.
- Glycine substitution hotspots mapped across **the entire collagen gene family**.

Note: This research was conducted independently as a self-directed project during my undergraduate studies, without formal supervision or external guidance.